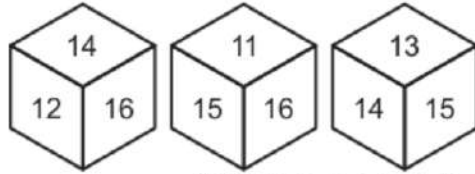


Q:1 Three different positions of the same dice are given below. Find the number opposite of 12.

नीचे एक ही पासे की तीन अलग-अलग स्थितियाँ दी गई हैं। 12 के विपरीत संख्या ज्ञात कीजिए।



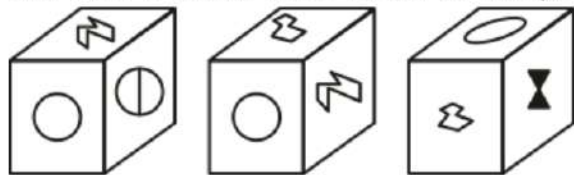
[DSSSB PGT English (Male) - 13 Feb 2024]

- Shift 1]

- | | |
|-------|-------|
| 1. 15 | 2. 16 |
| 3. 14 | 4. 11 |

Q:2 Three different position of the same dice are shown. Which pattern will be on the face opposite to the one having 'Ø'?

एक ही पासे की तीन अलग-अलग स्थितियाँ दर्शायी गई हैं। 'Ø' दर्शाने वाले फलक के विपरीत फलक पर कौन-सा पैटर्न होगा?

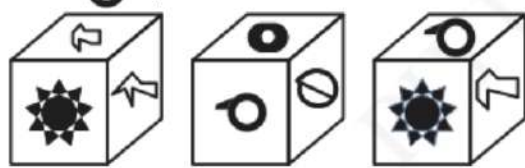


[DSSSB TGT CS - 24 June 2023 - Shift 3]

- | | |
|----|----|
| 1. | 2. |
| 3. | 4. |

Q:3 Three different positions of the same dice are shown. Which pattern will be on the face opposite to the one showing 'Ø'?

एक ही पासे की तीन अलग-अलग स्थितियाँ दिखाई गई हैं। कौन सा पैटर्न 'Ø' दर्शाने वाले फलक के विपरीत फलक पर होगा?

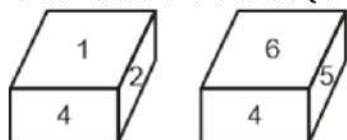


[DSSSB TGT - 24 June 2023 - Shift 2]

- | | |
|----|----|
| 1. | 2. |
| 3. | 4. |

Q:4 On the basis of the two figure orientations of the dice shown, which face is to the opposite of number 4?

दिखाए गए पासे की दो आकृति अभिविन्यासों के आधार पर, कौन सा फलक संख्या 4 के विपरीत है?



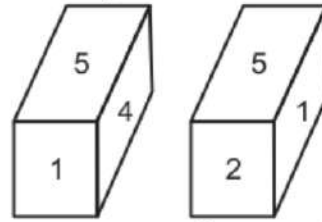
[DSSSB DOE PRT - 29 March 2022 - Shift

3]

- | | |
|------|------|
| 1. 1 | 2. 2 |
| 3. 5 | 4. 3 |

Q:5 From the following two different positions of a dice, which number lies to the opposite of 2?

एक पासे की निम्नलिखित दो भिन्न स्थितियों में, कौन सी संख्या 2 के विपरीत है?

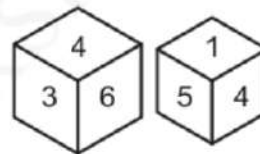


[DSSSB PRT (Assistant Teacher Primary) - 25 March 2022 - Shift 3]

- | | |
|------|------|
| 1. 4 | 2. 6 |
| 3. 1 | 4. 5 |

Q:6 In the following question, two positions of a die are given. Which number will appear on the face opposite to the face with the number 5?

निम्नलिखित प्रश्न में एक पासे की दो स्थितियाँ दी गई हैं। संख्या 5 वाले फलक के विपरीत फलक पर कौन सी संख्या आएगी?



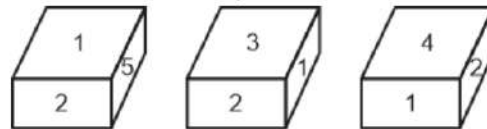
[DSSSB DOE PRT - 20 March 2022 - Shift

1]

- | | |
|------|------|
| 1. 1 | 2. 3 |
| 3. 6 | 4. 4 |

Q:7 On the basis of the three orientations of the dice shown, which face is to the opposite of number 2?

दिखाए गए पासे की तीन दिशाओं के आधार पर, कौन सा फलक संख्या 2 के विपरीत है?



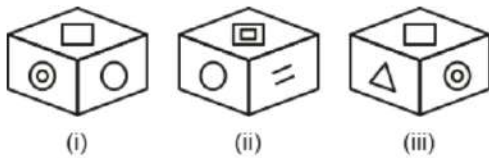
[DSSSB DOE PRT - 29 March 2022 - Shift

3]

- | | |
|------|------|
| 1. 3 | 2. 1 |
| 3. 5 | 4. 6 |

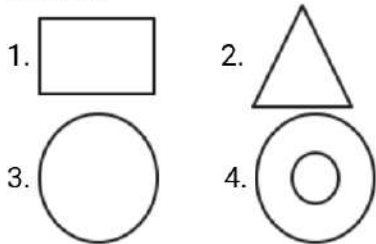
Q:8 Given below are three positions of the same dice. Which of the symbols in the options appears at the bottom in Figure (ii)?

नीचे एक ही पासे की तीन स्थितियाँ दी गई हैं। विकल्पों में से कौन सा प्रतीक आकृति (ii) में सबसे नीचे दिखाई देता है?



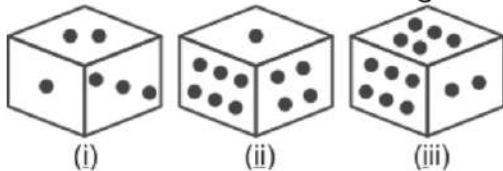
[DSSSB NDMC PRT - 26 March 2022 -

Shift 1]



Q:9 Three positions of the same dice are shown below, find the number of dots that lies opposite to the face with 6 dots.

नीचे एक ही पासे की तीन स्थितियाँ दर्शाई गई हैं, 6 बिंदुओं वाले फलक के विपरीत फलक पर स्थित बिंदुओं की संख्या ज्ञात कीजिए।



[DSSSB PRT (Assistant Teacher Primary)

- 26 March 2022 - Shift 2]

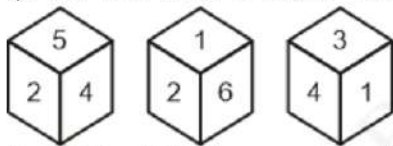
1. 2 2. 3
3. 1 4. 5

Q:10 A dice has six numbers marked 1, 2, 3, 4, 5, and 6 on its faces. Three positions of dice are shown below.

Which of the following statements is correct?

- I. 4 and 6 are on opposite faces.
II. 1 and 5 are on opposite faces.

एक पासे के फलकों पर 1, 2, 3, 4, 5 और 6 से अंकित छह संख्याएं हैं। नीचे पासों की तीन स्थितियाँ दर्शाई गई हैं।



निम्नलिखित में से कौन सा कथन सही है?

- I. 4 और 6 विपरीत फलकों पर हैं।
II. 1 और 5 विपरीत फलकों पर हैं।

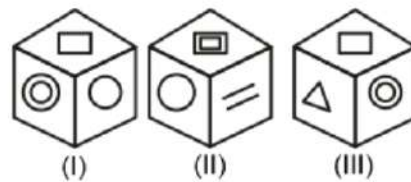
[DSSSB PRT (Assistant Teacher Primary)

- 24 March 2022 - Shift 3]

1. Only II / केवल II
2. Neither I nor II / न तो I और न ही II
3. Only I / केवल I
4. Both I and II / I और II दोनों

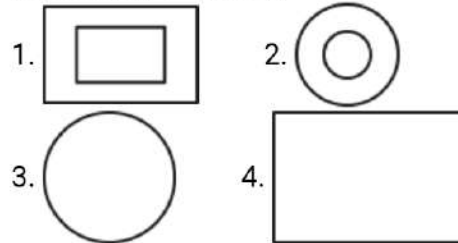
Q:11 Given below are three positions of the same dice. Which of the symbols in the options appears opposite the face having a triangle?

नीचे एक ही पासे की तीन स्थितियाँ दी गई हैं। विकल्पों में से कौन सा प्रतीक त्रिभुज वाले फलक के विपरीत दिखाई देता है?



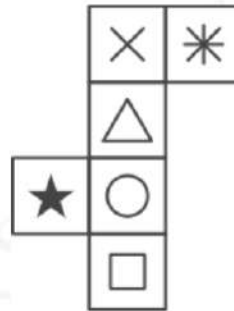
[DSSSB PRT (Assistant Teacher) - 27

March 2022 - Shift 1]



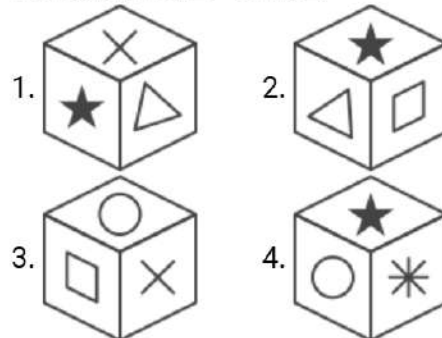
Q:12 Which of the following cubes in the answer figure can be made based on the unfolded cube in the question figure?

उत्तर आकृति में निम्नलिखित में से कौन सा घन प्रश्न आकृति में खुले घन के आधार पर बनाया जा सकता है?



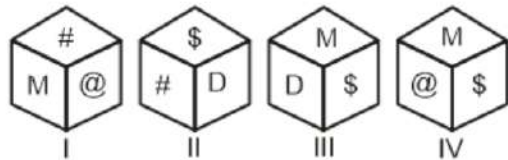
[DSSSB PRT (Assistant Teacher Primary) -

26 March 2022 - Shift 2]



Q:13 When the given paper is folded, which of the following cubes can be formed?

दिए गए कागज को मोड़ने पर, निम्नलिखित में से कौन सा घन बनाया जा सकता है?



[DSSSB DOE PRT - 07 March 2022 - Shift

2]

1. II and III / II तथा III
2. Only III / केवल III
3. I, II and III / I, II तथा III
4. I and II / I तथा II

Q:14 Two adjacent faces of a solid cube are painted with black colour. The faces opposite to the black faces is painted with blue colour while the remaining faces are painted with yellow colour. After painting, this cube has been divided into 125 equal cubes. How many cubes have no faces painted?

एक ठोस घन के दो आसन्न फलकों को काले रंग से रंगा गया है। काले फलकों के विपरीत फलकों को नीले रंग से रंगा गया है जबकि शेष फलकों को पीले रंग से रंगा गया है। रंगने के बाद, इस घन को 125 बराबर घनों में विभाजित किया गया है। कितने घनों के कोई फलक रंगे नहीं हैं?

[DSSSB PRT (Assistant Teacher Primary)

- 25 March 2022 - Shift 1]

1. 8
2. 35
3. 27
4. 24

Q:15 A cube of side 4 cm is painted green on all its faces and then divided into smaller cubes with every 2 cm sides. How many small cubes are obtained?

4 सेंमी भुजा वाले घन को उसकी सभी फलकों पर हरे रंग से रंगा गया है तथा फिर प्रत्येक 2 सेंमी. भुजा वाले छोटे घनों में विभाजित किया गया है। कितने छोटे घन प्राप्त हुए हैं?

[DSSSB DOE PRT - 07 March 2022 - Shift

2]

1. 4
2. 64
3. 27
4. 8

Q:16 Three positions of a dice are given below. If '6' is on the bottom face, then what can come on the top face?

नीचे एक पासे की तीन स्थितियाँ दी गई हैं। यदि नीचे वाले फलक पर '6' है, तो ऊपर वाले फलक पर क्या आएगा?



[DSSSB TGT SST (Female) - 09 Oct 2021 -

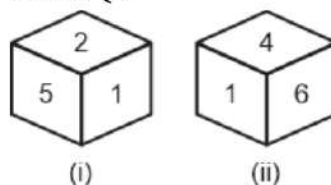
Shift 2]

1. 1
2. 2
3. 3
4. 4

Q:17 In the following question, a die has six faces 1, 2, 3, 4, 5 and 6. Two positions of the same cube are given.

Which number is opposite to number 6?

निम्नलिखित प्रश्न में, एक पासे के छह फलक 1, 2, 3, 4, 5 और 6 हैं। एक ही घन की दो स्थितियाँ दी गई हैं। कौन सी संख्या संख्या 6 के विपरीत है?



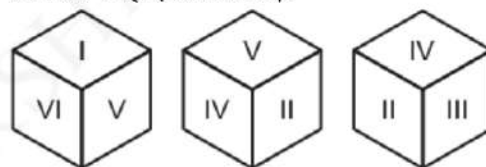
[DSSSB PGT Political Science (Male) - 04

July 2021]

1. 2
2. 5
3. 3
4. 6

Q:18 Three different positions of the same dice are given below. Find the symbol opposite of (III).

नीचे एक ही पासे की तीन अलग-अलग स्थितियाँ दी गई हैं। (III) के विपरीत चिह्न ज्ञात कीजिए।



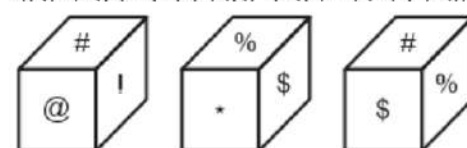
[DSSSB TGT Hindi (Female) - 05 Sep 2021

- Shift 1]

1. IV
2. V
3. VI
4. I

Q:19 Three different positions of the same dice are given below. Which symbol is on the face opposite to the face showing '@'?

एक ही पासे की तीन अलग-अलग स्थितियाँ नीचे दी गई हैं। '@' दर्शाने वाले फलक के विपरीत फलक पर कौन-सा चिह्न है?



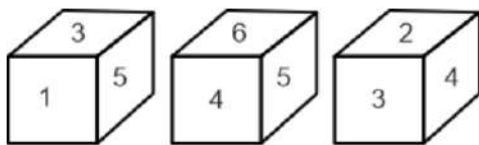
[DSSSB TGT Maths (Male) - 02 Sept 2021

- Shift 2]

1. #
2. %
3. !
4. \$

Q:20 Three different positions of the same dice are shown. What will be the number on the face opposite to the side having '1'?

एक ही पासे की तीन अलग-अलग स्थितियाँ दिखाई गई हैं। '1' वाले पक्ष के विपरीत फलक पर कौन सी संख्या होगी?



[DSSSB PGT Commerce (Male) - 28 June]

2021 - Shift 1]

1. 3 2. 6
3. 2 4. 4

Q:21 Three different positions of the same dice are given below. How many dots will be on the face opposite to the face bearing 3 dots?

एक ही पासे की तीन अलग-अलग स्थितियाँ नीचे दी गई हैं। 3 बिंदु वाले फलक के विपरीत फलक पर कितने बिंदु होंगे?



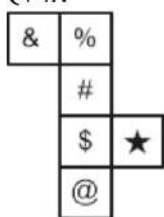
[DSSSB PGT English (Female) - 30 June]

2021 - Shift 2]

1. 1 2. 6
3. 2 4. 5

Q:22 A cube is made by folding the given sheet. In the cube so formed, which of the following pairs of symbols will be on opposite sides?

दी गई शीट को मोड़कर एक घन बनाया जाता है। इस प्रकार बने घन में, निम्नलिखित में से प्रतीकों का कौन सा युग्म विपरीत भुजाओं पर होगा?



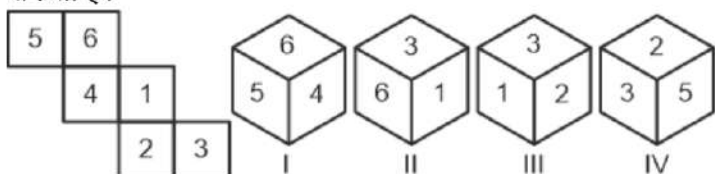
[DSSSB TGT Maths (Male) - 02 Sep 2021 -

Shift 1]

1. % and \$ / % और \$
2. # and * / # और *
3. @ and % / @ और %
4. & and \$ / & और \$

Q:23 By folding the given paper, which of the following cube can be made?

दिए गए कागज को मोड़कर निम्नलिखित में से कौन सा घन बनाया जा सकता है?



[DSSSB TGT SST (Female) - 09 Oct 2021 -

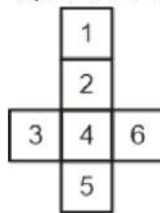
Shift 2]

1. I, II, III and IV / I, II, III और IV
2. I and III / I और III
3. I, II and IV / I, II और IV

4. I, II and II / I, II और II

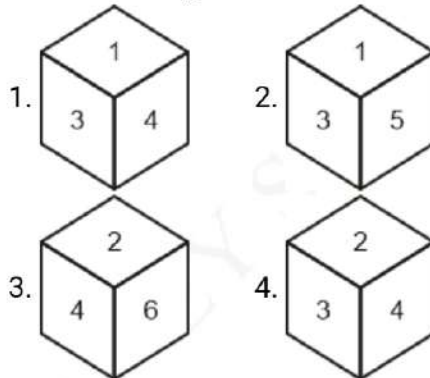
Q:24 Which of the following figures cannot be formed on folding the given cube in the question?

प्रश्न में दिए गए घन को मोड़ने पर निम्नलिखित में से कौन सी आकृति नहीं बन सकती है?



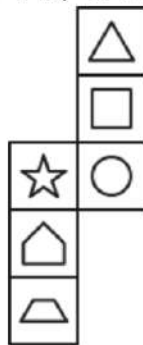
[DSSSB PGT Chemistry (Female) - 18 July]

2021 - Shift 1]



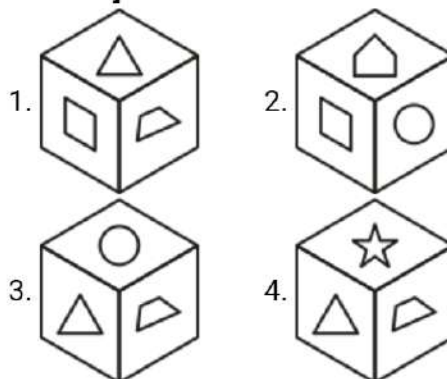
Q:25 Which of the following figures can be formed by folding the given cube in the question?

प्रश्न में दिए गए घन को मोड़कर निम्नलिखित में से कौन सी आकृति बनाई जा सकती है?



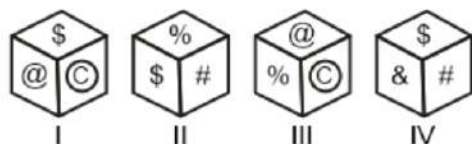
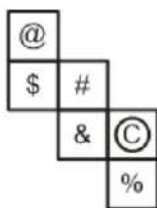
[DSSSB TGT Hindi (Female) - 04 Sep 2021]

- Shift 3]



Q:26 By folding the given paper, which of the following cube can be made?

दिए गए कागज को मोड़कर निम्नलिखित में से कौन सा घन बनाया जा सकता है?



[DSSSB TGT English (Female) - 14 Sep 2021 - Shift 3]

Q:27 Pairs of adjacent sides of a big cube are coloured with red, yellow and brown colour. Now this big cube is divided into 64 small equal cubes. How many small cubes will have 3 faces painted with different colours?

1. I, II, III and IV / I, II, III और IV
2. I and II / I और II
3. III and IV / III और IV
4. II, III and IV / II, III और IV

एक बड़े घन की आसन्न भुजाओं के युग्म लाल, पीले और भूरे रंग से रंगे गए हैं। अब यह बड़ा घन, 64 छोटे बराबर घनों में विभाजित हो गया है। कितने छोटे घनों के 3 फलक अलग-अलग रंगों से रंगे होंगे?

[DSSSB TGT CS - 07 Aug 2021 - Shift 1]

1. 8
2. 0
3. 2
4. 6

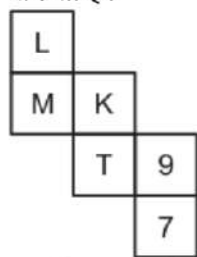
Q:28 A cube is taken and 3 equal cuts are made on each of the axis. How many small cubes will be obtained? एक घन लिया जाता है और प्रत्येक अक्ष पर 3 बराबर कट लगाए जाते हैं। कितने छोटे घन प्राप्त होंगे?

[DSSSB TGT English (Female) - 14 Sep 2021 - Shift 3]

1. 9
2. 16
3. 64
4. 125

Q:29 By folding the given paper, which of the following cubes can be made?

दिए गए कागज को मोड़कर निम्नलिखित में से कौन सा घन बनाया जा सकता है?

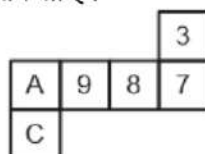


[DSSSB PRT - 14 Nov 2019 - Shift 1]

1. Both I and III / I और III दोनों
2. Only I / केवल I
3. Both II and III / II और III दोनों
4. Only III / केवल III

Q:30 When the given paper is folded, which of the following cubes can be formed?

दिए गए कागज को मोड़ने पर, निम्नलिखित में से कौन सा घन बनाया जा सकता है?

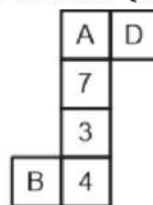


[DSSSB PRT - 15 Nov 2019 - Shift 1]

1. Only I / केवल I
2. II & III Both / II तथा III दोनों
3. I & III Both / I तथा III दोनों
4. Only III / केवल III

Q:31 By folding the given paper, which of the following cubes cannot be made?

दिए गए कागज को मोड़कर निम्नलिखित में से कौन सा घन नहीं बनाया जा सकता है?



[DSSSB PRT - 13 Nov 2019 - Shift 1]

1. Both I and II / I और II दोनों
2. Both II and III / II और III दोनों
3. Only II / केवल II
4. Only III / केवल III

Q:32 A big solid cube is painted with red colour on two opposite faces, yellow colour on two adjacent faces and green colour on the remaining two faces. Now 3 cuts are made on each axis of this cube and then some small cubes are obtained. How many small cubes will have one face painted with red colour and one face painted with yellow colour?

एक बड़े ठोस घन के दो विपरीत फलकों पर लाल रंग, दो आसन्न फलकों पर पीला रंग तथा शेष दो फलकों पर हरा रंग लगाया गया है। अब इस घन के प्रत्येक अक्ष पर 3 कट लगाए जाते हैं तथा कुछ छोटे घन प्राप्त होते हैं। कितने छोटे घनों के एक फलक पर लाल रंग तथा एक फलक पर पीला रंग लगाया गया होगा?

[DSSSB PRT - 13 Nov 2019 - Shift 1]

- | | |
|-------|-------|
| 1. 20 | 2. 16 |
| 3. 14 | 4. 12 |

Q:33 A cube of side 8 cm is painted red on all the sides. It is then cut into smaller cubes, each of side 2 cm. How many of these smaller cubes have one or more sides painted red?

8 सेमी भुजा वाले एक घन को सभी भुजाओं पर लाल रंग से रंगा गया है। फिर इसे 2 सेमी भुजा वाले छोटे घनों में काटा गया है। इन छोटे घनों में से कितने की एक या अधिक भुजाओं पर लाल रंग से रंगा गया है?

[KVS PRT - 2018]

- | | |
|-------|-------|
| 1. 32 | 2. 48 |
| 3. 56 | 4. 24 |

Q:34 A solid brick of Gold measuring 18 cm × 48 cm × 16 cm is melted and converted into a cubical shape. An edge of this new shape is equal to:

18 सेमी × 48 सेमी × 16 सेमी माप वाली सोने की एक ठोस ईंट को पिघलाकर एक घनाकार आकार में परिवर्तित किया जाता है। इस नए आकार का एक किनारा बराबर है:

[KVS PRT - 2018]

1. 24 cm / 24 सेमी
2. 12 cm / 12 सेमी
3. 18 cm / 18 सेमी
4. 48 cm / 48 सेमी

Q:35 A cube is to be coloured in such a way as to avoid the same colour on adjacent surfaces. What is the minimum number of colours you will require?

एक घन को इस तरह से रंगा जाना चाहिए कि आसन्न सतहों पर एक ही रंग न हो। आपको न्यूनतम कितने रंगों की आवश्यकता होगी?

[KVS PRT - 2015]

1. Six / छह
2. Four / चार
3. Three / तीन
4. Nine / नौ

Answer Key

1. (1)	2. (3)	3. (4)	4. (4)	5. (1)
6. (3)	7. (4)	8. (4)	9. (2)	10. (4)
11. (3)	12. (1)	13. (2)	14. (3)	15. (4)
16. (4)	17. (1)	18. (2)	19. (4)	20. (4)
21. (2)	22. (1)	23. (1)	24. (1)	25. (1)
26. (3)	27. (3)	28. (3)	29. (4)	30. (1)
31. (4)	32. (3)	33. (3)	34. (1)	35. (3)

Answers and Solutions

Q:1 The correct answer is **Option 1** i.e. **15**.

In the first and third dice, 14 is common. So, start rotating dice in the clockwise direction.

Clockwise direction:

Dice 1	14	16	12
Dice 3	14	13	15

So, 12 is opposite to 15.

Q:2 The correct answer is **Option 3**

On comparing dice 1 and dice 2 we get that they have two same figures. So, the remaining unique figures will be opposite to each other.

So,  is opposite to .

Q:3 The correct answer is **Option 4** i.e. 

According to the given dice, opposite pairs will be :

Dice 2			
Dice 3			

Q:4 The correct answer is **Option 4** i.e. **3**.

On comparing dice 1 and dice 2 we get that in both dice 4 is in the same place. So, 1, 2, 5, and 6 are opposite of 4. The remaining number **3** will be opposite to **4**.

Q:5 The correct answer is **Option 1** i.e. **4**.

On comparing both dice we get that they have two same numbers. So, the remaining numbers will be opposite to each other.

So, 2 is opposite to 4.

Q:6 The correct answer is **Option 3** i.e. **6**.

In this case, 4 is common in both the dices, so considering the numbers in the clockwise direction from 4, we get the numbers as,

Dice 1	4	6	3
Dice 2	4	5	1

Now comparing both the dice we find that 6 is opposite to 5, and 3 is opposite to 1.

Q:7 The correct answer is **Option 4** i.e. **6**.

On comparing dice 1, 2, and 3 we get that 1, 3, 4, and 5 numbers are not the opposite of 2. So, the remaining number **6** will be opposite to **2**.

Q:8 The correct answer is **Option 4**

From dice 1 and 2 we get, a circle as a common part so we will show each figure on both sides in the clockwise direction of the circle.

	Common Figure	Opposite Figure	Opposite Figure
Dice 1			
Dice 2			

Hence, the **double circle** in the options appears at the bottom in Figure (ii) as it is opposite to the **double rectangle**.

Q:9 The correct answer is **Option 2** i.e. **3**.

When comparing dice 2 and dice 3, 1, 4, 5 and 2 numbers are adjacent to 6. So, the remaining number 3 will be opposite to 6.

Q:10 The correct answer is **Option 4** i.e. **Both I and II**.

II. On comparing dice 1 and dice 2, from the number 2 in a clockwise direction we get

Dices	Common number	Opposite number	Opposite number
Dice 1	2	5	4
Dice 2	2	1	6

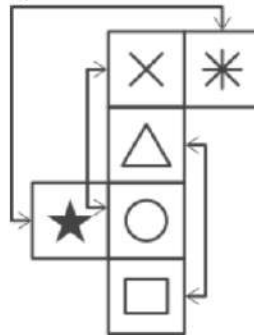
I. 4 and 6 are on opposite faces: **Correct**

II. 1 and 5 are on opposite faces: **Correct**

Q:11 The correct answer is **Option 3**

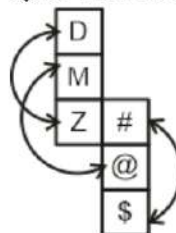
On comparing dice 1 and dice 3 we get that they have two same figures. So, the remaining unique figures will be opposite to each other: circle and triangle. Therefore, Circle will be opposite to the triangle.

Q:12 The correct answer is **Option 1**



To form a cube the opposite pairs should not be adjacent. So, **Option (1)** can be formed.

Q:13 The correct answer is **Option 2** i.e. **Only III**



Opposite pairs = D $\Leftrightarrow$ Z, # $\Leftrightarrow$ \$, and M $\Leftrightarrow$ @
So, only III can be formed.

Q:14 The correct answer is **Option 3** i.e. **27**.

There are 125 small cubes in total.

The number of cubes having 3 surfaces colored = 8

The number of cubes having 2 surfaces colored, which are on the edges = 3×12 (edges) = 36.

The number of cubes having 1 surface colored = 9×6 (faces) = 54.

The number of cubes with no surface colored = 125 minus $(8 + 36 + 54) = 27$

Q:15 The correct answer is **Option 4** i.e. **8**

Side of larger cube = $S_1 = 4\text{cm}$

Side of smaller cube = $S_2 = 2\text{ cm}$

The total number of smaller cubes obtained = (Volume of larger cube/Volume of smaller cube)

$$= (S_1)^3 / (S_2)^3$$

$$= (4 \times 4 \times 4) / (2 \times 2 \times 2)$$

$$= 64 / 8$$

$$= 8$$

Q:16 The correct answer is **Option 4** i.e. **4**

From the figure: 1, 2, 3, 5 are adjacent to 6. Hence, 4 is opposite to 6.

Q:17 The correct answer is **Option 1** i.e. **2**.

As 1 is common in both dice, we write all dice numbers in the clockwise direction from 1.

Dice 1	1	5	2
Dice 2	1	4	6

Hence opposite of 6 is 2.

Q:18 The correct answer is **Option 2** i.e. **V**.

In the second and third cubes, IV and II are common. So, the remaining faces are opposite to each other. So, III is opposite to V.

Q:19 The correct answer is **Option 4** i.e. **\$**.

In the first and third dice, # is common. So, start rotating dice in the clockwise direction.

Clockwise direction:

Dice 1	#	!	@
Dice 2	#	%	\$

So, @ is opposite to \$.

Q:20 The correct answer is **Option 4** i.e. **4**.

Logic: Take the common number from dice 1 and 2 and then write the number in their clockwise direction.

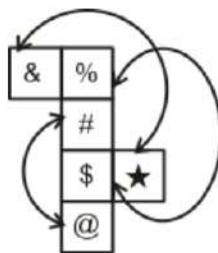
Dice1	5	1	3
Dice 2	5	4	6

So, 4 is opposite to number 1.

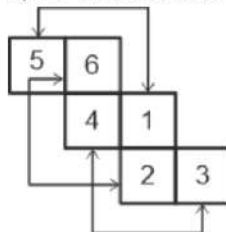
Q:21 The correct answer is **Option 2** i.e. **6**.

In all the 3 dice, 3 is common and the numbers 4, 2, 5, and 1 are adjacent to 3. So, 6 will be opposite to 3.

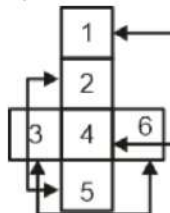
Q:22 The correct answer is **Option 1** i.e. **% and \$**.



Q:23 The correct answer is **Option 1** i.e. **I, II, III and IV**.



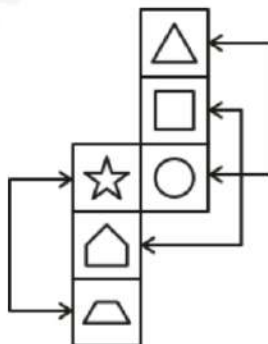
Q:24 The correct answer is **Option 1**



To form a cube the opposite pairs should not be adjacent. In option (1), 1 and 4 are opposite pairs which are shown adjacent. So, it can't be formed.

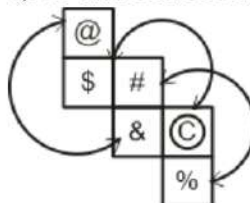
Q:25 The correct answer is **Option 1**

After folding the paper, pair of opposite sides will be as shown below:



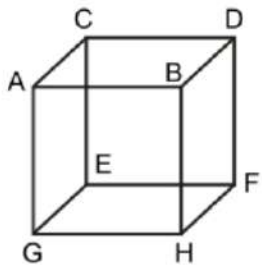
Only option 1 fulfills this.

Q:26 The correct answer is **Option 3** i.e. **III and IV**



To form a cube the opposite pairs should not be adjacent. In Figures I and II, \$ and ©, #, and % are opposite pairs shown adjacent. So, these can't be formed. But III and IV can be formed.

Q:27 The correct answer is **option 3** i.e. **2**



Let faces **ABHG** and **BDFH** be Red color, faces **ACDB** and **CDFE** be Yellow color, and faces **ACEG** and **EFHG** be Brown color.

The common vertices **A** and **F** will have three faces of different colors.

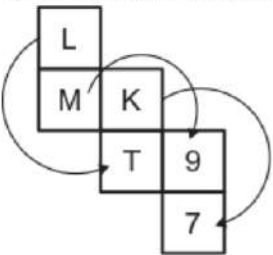
Q:28 The correct answer is **Option 3** i.e. **64**

For a cube, if we make 3 cuts along each axis (X, Y, and Z),

The total number of smaller cubes will be: $(X + 1) \times (Y + 1) \times (Z + 1)$ {we are essentially dividing each side into 4 equal parts.}

$$4 \times 4 \times 4 = 64$$

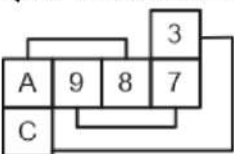
Q:29 The correct answer is **Option 4** i.e. **Only III**.



To form a cube the opposite pairs should not be adjacent.

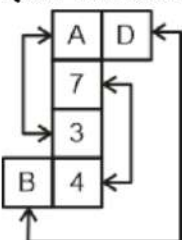
In **Figure I**, L and T are opposite pairs; in **Figure II**, K and 7 are opposite pairs shown adjacent. So, only **Figure III** can be formed.

Q:30 The correct answer is **Option 1** i.e. **Only I**.



Two opposite pairs can't be adjacent. So, dice II and III can't be formed.

Q:31 The correct answer is **Option 4** i.e. **Only III**.



In III, opposite pairs 7 and 4 are adjacent to each other. So, this cannot be formed.

Q:32 The correct answer is **Option 3** i.e. **14**.

A big solid cube is cut into small cubes.

If 3 cuts are made on each axis then the total cubes made from each edge is 4.

$$\text{Now the total number of cubes} = 4^3 = 64$$

The number of cubes will have one face painted red colour and one face painted yellow colour the cubes which are combined faces yellow and red coloured at the edges = $4 \times 4 - 2$ (cubes which also have green faces) $16 - 2 = 14$

Q:33 The correct answer is **option 3** i.e. **56**

$$\text{Large cube side length} = S = 8 \text{ cm}$$

$$\text{Small cube side length} = s = 2 \text{ cm}$$

$$\text{Number of small cubes per edge} = S/2 = 8/2 = 4$$

$$\text{Total number of small cubes} = (4)^3 = 64$$

$$\text{Cubes at corner} = 8$$

$$\text{Cubes at edge (excluding corner)} = 12 \times (n - 2) = 12 \times (4 - 2) = 24$$

$$\text{Face-only cubes (excluding edges and corners)} = 6 \times (n - 2)^2 = 6 \times (4 - 2)^2 = 24$$

$$\text{Total cubes having at least one side painted red} = 8 + 24 + 24 = 56$$

Q:34 The correct answer is **option 1** i.e. **24 cm**

$$\text{Volume of original rectangular brick} = 18 \text{ cm} \times 48 \text{ cm} \times 16 \text{ cm} = 13824 \text{ cm}^3$$

Let the length of the new cubical shape = x

Edge of the new cubical shape

$$x^3 = 13824$$

$$x = \sqrt[3]{13824}$$

$$x = 24 \text{ cm}$$

Q:35 The correct answer is **option 3** i.e. **3**

As, no two adjacent surfaces can same colour, any pair of two opposite faces can have same colour.

$$\text{Minimum numbers of colours you will be require} = 6 \div 2 = 3$$

Hence, the correct answer is '**Three**'.